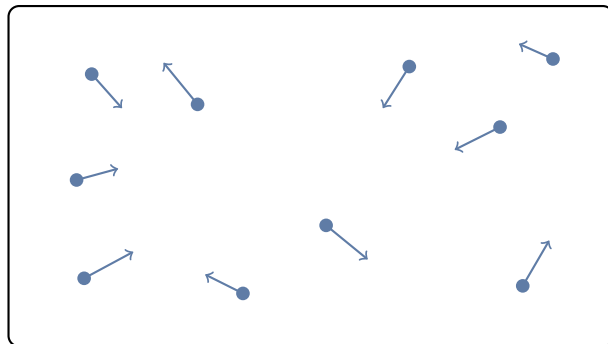


Lecture Notes on Ideal Gas Theory

From the Ideal Gas Law to Kinetic Theory and Thermodynamic Processes



molecules in random motion

Designed as practical mathematical physics notes

May 17, 2026

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1 What is an ideal gas?

An **ideal gas** is a simplified model of a gas in which the molecules are treated as point particles that move randomly and collide elastically. The model ignores molecular volume and long-range intermolecular forces. Despite its simplicity, it describes many real gases very well when the gas is sufficiently dilute and not close to condensation.

Main idea. Ideal gas theory connects a macroscopic description of matter, such as pressure, volume and temperature, with a microscopic description based on molecular motion.

The central formula is the **ideal gas equation**

$$pV = nRT. \quad (1)$$

Here

- p is pressure, measured in pascals: $\text{Pa} = \text{N}/\text{m}^2$,
- V is volume, measured in cubic metres: m^3 ,
- n is the amount of substance, measured in moles,
- R is the universal gas constant,
- T is absolute temperature, measured in kelvin.

The same equation can also be written as

$$pV = Nk_B T, \quad (2)$$

where N is the number of molecules and k_B is the Boltzmann constant. The two constants are related by

$$R = N_A k_B, \quad (3)$$

where N_A is Avogadro's number.

1.1 State variables

A gas in thermodynamic equilibrium can be described by a small number of macroscopic variables. The most important ones are:

Quantity	Symbol	Typical SI unit
Pressure	p	Pa
Volume	V	m^3
Temperature	T	K
Amount of substance	n	mol
Number of molecules	N	dimensionless
Internal energy	U	J

In ideal gas formulas, temperature must be in kelvin, not in degrees Celsius. The conversion is

$$T[\text{K}] = t[^\circ\text{C}] + 273.15.$$

1.2 Why the model is useful

The ideal gas model is useful because it is simple, quantitative and predictive. It allows us to calculate how pressure changes when a gas is compressed, how much work a gas does during expansion, how temperature is related to molecular kinetic energy, and how heat capacities determine thermodynamic processes.

2 Empirical gas laws

Before introducing kinetic theory, it is useful to understand how the ideal gas law summarizes several empirical laws.

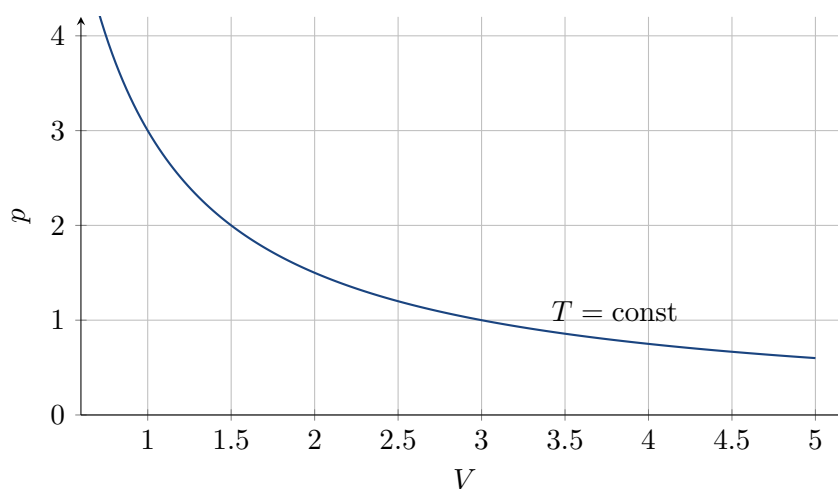
2.1 Boyle's law

For a fixed amount of gas at constant temperature,

$$pV = \text{constant}. \quad (4)$$

Thus pressure is inversely proportional to volume:

$$p \propto \frac{1}{V}. \quad (5)$$



2.2 Charles's law

For a fixed amount of gas at constant pressure,

$$\frac{V}{T} = \text{constant}, \quad (6)$$

so volume is proportional to absolute temperature:

$$V \propto T. \quad (7)$$

2.3 Gay-Lussac's law

For a fixed amount of gas at constant volume,

$$\frac{p}{T} = \text{constant}, \quad (8)$$

so pressure is proportional to absolute temperature:

$$p \propto T. \quad (9)$$

2.4 Combining the laws

For a fixed amount of gas, the three empirical relations combine into

$$\frac{pV}{T} = \text{constant}. \quad (10)$$

For n moles this constant equals nR , giving the ideal gas equation:

$$pV = nRT. \quad (11)$$

Example 1. A gas occupies $V_1 = 2.0$ L at pressure $p_1 = 100$ kPa. It is compressed isothermally to $V_2 = 0.50$ L. Find p_2 .

Since $pV = \text{constant}$,

$$p_1 V_1 = p_2 V_2, \quad p_2 = \frac{p_1 V_1}{V_2} = \frac{100 \text{ kPa} \cdot 2.0 \text{ L}}{0.50 \text{ L}} = 400 \text{ kPa}.$$

3 Using the ideal gas equation

The equation

$$pV = nRT \quad (12)$$

can be rearranged depending on the unknown quantity:

$$p = \frac{nRT}{V}, \quad (13)$$

$$V = \frac{nRT}{p}, \quad (14)$$

$$n = \frac{pV}{RT}, \quad (15)$$

$$T = \frac{pV}{nR}. \quad (16)$$

3.1 Number of molecules

If the number of molecules is used instead of the amount in moles, then

$$pV = Nk_B T. \quad (17)$$

The number of molecules is related to the amount of substance by

$$N = nN_A. \quad (18)$$

3.2 Units

The product pV has units of energy:

$$[pV] = \text{Pa m}^3 = \frac{\text{N}}{\text{m}^2} \text{m}^3 = \text{N m} = \text{J}. \quad (19)$$

Therefore nRT and $Nk_B T$ also have units of energy.

Useful constants:

$$R \approx 8.314 \text{ J/(mol K)}, \quad k_B \approx 1.381 \times 10^{-23} \text{ J/K}, \quad N_A \approx 6.022 \times 10^{23} \text{ mol}^{-1}.$$

Example 2. How many moles are in a vessel of volume 0.010 m^3 at pressure $2.0 \times 10^5 \text{ Pa}$ and temperature 300 K ?

$$n = \frac{pV}{RT} = \frac{(2.0 \times 10^5 \text{ Pa})(0.010 \text{ m}^3)}{(8.314 \text{ J/(mol K)})(300 \text{ K})} \approx 0.80 \text{ mol.}$$

4 Pressure, volume and work

In thermodynamics, the gas can exchange energy with its surroundings through **heat** and **work**. If a gas expands against an external pressure, it does work.

For a quasistatic process, the infinitesimal work done by the gas is

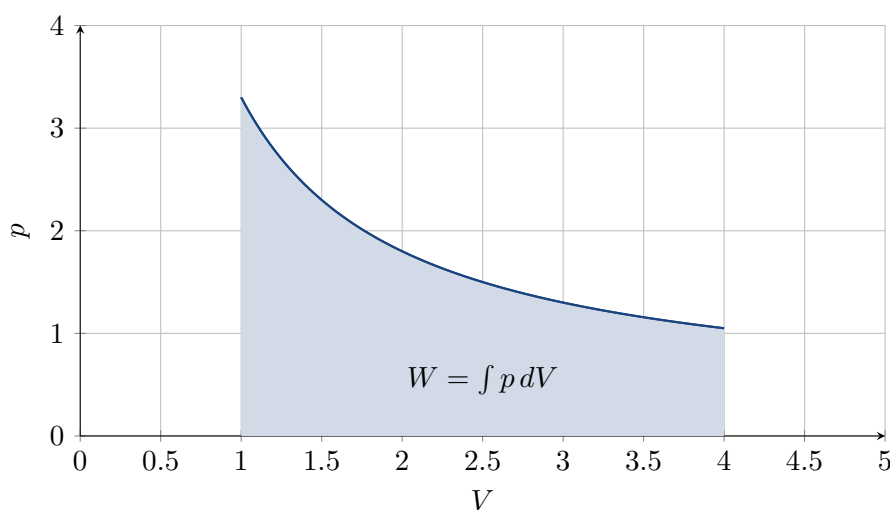
$$dW = p dV. \quad (20)$$

Hence the total work done by the gas is

$$W = \int_{V_i}^{V_f} p(V) dV. \quad (21)$$

4.1 Geometric interpretation

On a pV diagram, work is the area under the curve.



4.2 Work in common processes

Process	Condition	Work done by gas
Isochoric	$V = \text{const}$	$W = 0$
Isobaric	$p = \text{const}$	$W = p(V_f - V_i)$
Isothermal ideal gas	$T = \text{const}$	$W = nRT \ln(V_f/V_i)$
Adiabatic reversible	$Q = 0$	$W = (p_i V_i - p_f V_f)/(\gamma - 1)$

Sign conventions differ between textbooks. In these notes, W means work done by the gas. Then expansion gives $W > 0$, compression gives $W < 0$.

5 Microscopic assumptions of kinetic theory

Kinetic theory explains ideal gas behavior from molecular motion. The basic assumptions are:

1. The gas contains a very large number of molecules.
2. Molecules move randomly in all directions.
3. The size of molecules is negligible compared with the average distance between them.
4. There are no intermolecular forces except during collisions.
5. Collisions between molecules and with the walls are elastic.
6. The gas is in equilibrium, so statistical averages are well-defined.

5.1 Pressure from molecular collisions

Consider molecules in a cubic container. When molecules collide with the walls, they transfer momentum. The total rate of momentum transfer per unit area is the pressure.

For an ideal gas, kinetic theory gives

$$pV = \frac{1}{3}Nm\langle v^2 \rangle, \quad (22)$$

where m is the mass of one molecule and $\langle v^2 \rangle$ is the average of the squared molecular speed.

Comparing this with

$$pV = Nk_B T \quad (23)$$

gives the important relation

$$\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T. \quad (24)$$

Physical meaning of temperature. In an ideal gas, absolute temperature is proportional to the average translational kinetic energy of a molecule:

$$\langle K_{\text{trans}} \rangle = \frac{3}{2}k_B T.$$

6 Molecular speeds

The **root-mean-square speed** is defined by

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle}. \quad (25)$$

Using the kinetic theory relation,

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}. \quad (26)$$

If molar mass M is used, then $m = M/N_A$ and

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}. \quad (27)$$

Here M must be expressed in kg/mol.

Example 3. Estimate the rms speed of nitrogen molecules at $T = 300 \text{ K}$. Use $M = 0.028 \text{ kg/mol}$.

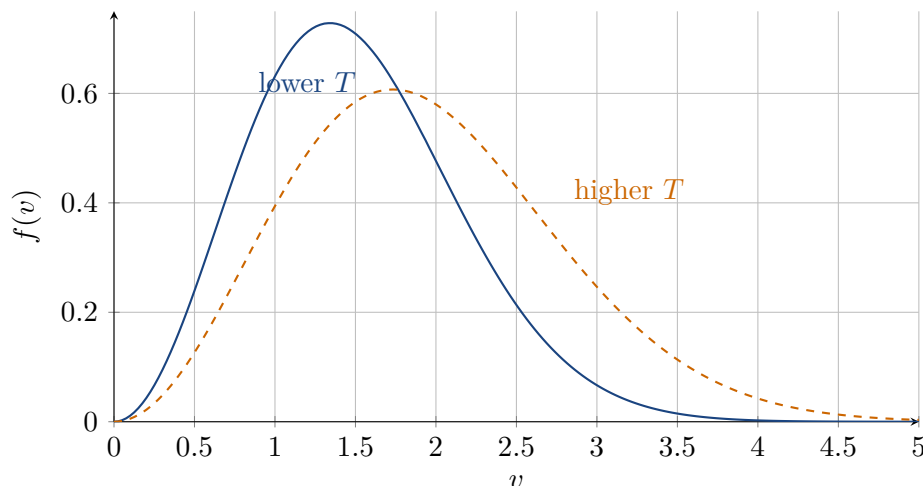
$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314 \text{ J/(mol K)})(300 \text{ K})}{0.028 \text{ kg/mol}}} \approx 517 \text{ m/s}.$$

6.1 Maxwell-Boltzmann distribution

Molecules in a gas do not all move with the same speed. At equilibrium, their speeds are described by the Maxwell-Boltzmann distribution. The exact formula is

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp \left(-\frac{mv^2}{2k_B T} \right), \quad v \geq 0. \quad (28)$$

This is a probability density for molecular speed.



As temperature increases, the distribution becomes broader and its maximum shifts toward larger speeds.

7 Internal energy

For a monatomic ideal gas, the only relevant microscopic energy is translational kinetic energy. Therefore

$$U = N \cdot \frac{3}{2} k_B T = \frac{3}{2} N k_B T. \quad (29)$$

Using $Nk_B = nR$,

$$U = \frac{3}{2} nRT. \quad (30)$$

More generally, for an ideal gas with f active quadratic degrees of freedom,

$$U = \frac{f}{2} nRT. \quad (31)$$

The parameter f depends on molecular structure and temperature.

Gas type	Typical active degrees of freedom	Approximate internal energy
Monatomic	$f = 3$	$U = \frac{3}{2} nRT$
Diatomic, moderate T	$f \approx 5$	$U \approx \frac{5}{2} nRT$
Nonlinear polyatomic	$f \approx 6$	$U \approx 3nRT$

The statement $U = \frac{3}{2} nRT$ is not universal. It applies to a monatomic ideal gas. For diatomic or polyatomic gases, rotational and vibrational degrees of freedom can matter.

8 The first law of thermodynamics

The first law of thermodynamics is an energy balance:

$$\Delta U = Q - W. \quad (32)$$

Here Q is heat supplied to the gas and W is work done by the gas.

Interpretation. Heat added to a gas can increase its internal energy, be converted into work, or both.

For an ideal gas, internal energy depends only on temperature:

$$\Delta U = nC_V\Delta T. \quad (33)$$

For a monatomic ideal gas,

$$C_V = \frac{3}{2}R, \quad \Delta U = \frac{3}{2}nR\Delta T. \quad (34)$$

9 Heat capacities

The molar heat capacity at constant volume is

$$C_V = \frac{1}{n} \left(\frac{\partial Q}{\partial T} \right)_V. \quad (35)$$

The molar heat capacity at constant pressure is

$$C_P = \frac{1}{n} \left(\frac{\partial Q}{\partial T} \right)_P. \quad (36)$$

For an ideal gas,

$$C_P - C_V = R. \quad (37)$$

The ratio

$$\gamma = \frac{C_P}{C_V} \quad (38)$$

is very important in adiabatic processes and sound waves.

For a monatomic ideal gas,

$$C_V = \frac{3}{2}R, \quad C_P = \frac{5}{2}R, \quad \gamma = \frac{5}{3}. \quad (39)$$

For a typical diatomic gas at room temperature,

$$C_V \approx \frac{5}{2}R, \quad C_P \approx \frac{7}{2}R, \quad \gamma \approx \frac{7}{5}. \quad (40)$$

10 Standard thermodynamic processes

10.1 Isochoric process

An isochoric process occurs at constant volume:

$$V = \text{constant}. \quad (41)$$

Since $dV = 0$, the work is zero:

$$W = 0. \quad (42)$$

The first law gives

$$Q = \Delta U = nC_V\Delta T. \quad (43)$$

10.2 Isobaric process

An isobaric process occurs at constant pressure:

$$p = \text{constant}. \quad (44)$$

The work is

$$W = p\Delta V. \quad (45)$$

Using $pV = nRT$, we also have

$$p\Delta V = nR\Delta T. \quad (46)$$

Therefore heat supplied is

$$Q = \Delta U + W = nC_V\Delta T + nR\Delta T = nC_P\Delta T. \quad (47)$$

10.3 Isothermal process

An isothermal process occurs at constant temperature:

$$T = \text{constant}. \quad (48)$$

For an ideal gas, U depends only on T , so

$$\Delta U = 0. \quad (49)$$

The first law becomes

$$Q = W. \quad (50)$$

The pressure is

$$p = \frac{nRT}{V}, \quad (51)$$

so the work is

$$W = \int_{V_i}^{V_f} \frac{nRT}{V} dV = nRT \ln \left(\frac{V_f}{V_i} \right). \quad (52)$$

10.4 Adiabatic process

An adiabatic process occurs without heat exchange:

$$Q = 0. \quad (53)$$

For a reversible adiabatic process of an ideal gas,

$$pV^\gamma = \text{constant}, \quad (54)$$

where $\gamma = C_P/C_V$. Equivalent forms are

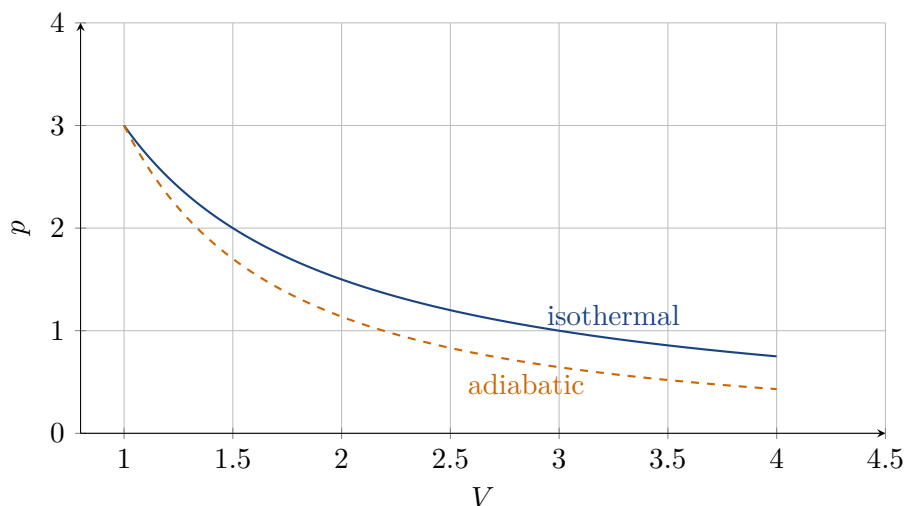
$$TV^{\gamma-1} = \text{constant}, \quad (55)$$

$$T^\gamma p^{1-\gamma} = \text{constant}. \quad (56)$$

Since $Q = 0$, the first law gives

$$\Delta U = -W. \quad (57)$$

For expansion, $W > 0$, so $\Delta U < 0$ and the gas cools down.



11 Adiabatic relations: a short derivation

For a reversible adiabatic process, $Q = 0$, so

$$dU = -p dV. \quad (58)$$

For an ideal gas,

$$dU = nC_V dT. \quad (59)$$

Thus

$$nC_V dT = -p dV. \quad (60)$$

Using $p = nRT/V$,

$$nC_V dT = -\frac{nRT}{V} dV. \quad (61)$$

After cancelling n and dividing by T ,

$$C_V \frac{dT}{T} = -R \frac{dV}{V}. \quad (62)$$

Because $R = C_P - C_V = (\gamma - 1)C_V$,

$$\frac{dT}{T} = -(\gamma - 1) \frac{dV}{V}. \quad (63)$$

Integrating gives

$$\ln T + (\gamma - 1) \ln V = \text{constant}, \quad (64)$$

which implies

$$TV^{\gamma-1} = \text{constant}. \quad (65)$$

Combining this with $pV = nRT$ gives

$$pV^\gamma = \text{constant}. \quad (66)$$

12 Sound speed in an ideal gas

Sound is a pressure wave. In a gas, sound propagation is approximately adiabatic because compressions and rarefactions happen quickly, with little time for heat exchange.

The speed of sound in an ideal gas is

$$c = \sqrt{\frac{\gamma p}{\rho}}, \quad (67)$$

where ρ is the mass density. Using $pV = nRT$ and $\rho = m_{\text{gas}}/V = nM/V$, one obtains

$$c = \sqrt{\frac{\gamma RT}{M}}. \quad (68)$$

Example 4. Estimate the speed of sound in air at $T = 293 \text{ K}$, using $\gamma = 1.40$ and $M = 0.029 \text{ kg/mol}$.

$$c = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{(1.40)(8.314 \text{ J/(mol K)})(293 \text{ K})}{0.029 \text{ kg/mol}}} \approx 343 \text{ m/s}.$$

13 Mean free path and collisions

In a real gas, molecules collide with one another. The **mean free path** is the average distance a molecule travels between collisions. A simple estimate is

$$\ell = \frac{1}{\sqrt{2} n_V \sigma}, \quad (69)$$

where $n_V = N/V$ is number density and σ is the collision cross-section.

Although the ideal gas equation ignores intermolecular forces, collisions are still allowed in kinetic theory because elastic collisions redistribute energy and maintain equilibrium.

Important distinction. The ideal gas model does not mean molecules never collide. It means interactions are negligible except during brief elastic collisions.

14 When the ideal gas model fails

The ideal gas approximation works best when:

- density is low,
- pressure is not too high,
- temperature is high compared with condensation temperature,
- intermolecular forces are weak or rarely important.

It becomes less accurate when:

- the gas is compressed strongly,
- the gas is close to liquefaction,
- molecules have significant attractive or repulsive interactions,
- molecular size cannot be neglected.

A common improved model is the van der Waals equation:

$$\left(p + a \frac{n^2}{V^2}\right)(V - nb) = nRT. \quad (70)$$

The parameter a accounts for attractive interactions, while b accounts for excluded molecular volume.

15 Common mistakes

1. **Using Celsius instead of kelvin.** Ideal gas formulas require absolute temperature.
2. **Mixing litres and cubic metres.** In SI calculations, use $1 \text{ L} = 1 \times 10^{-3} \text{ m}^3$.
3. **Using molar mass in grams per mole.** In formulas such as $v_{\text{rms}} = \sqrt{3RT/M}$, use M in kg/mol.
4. **Confusing heat and temperature.** Heat is energy transfer; temperature measures average molecular kinetic energy.
5. **Assuming all processes are isothermal.** A gas can be isothermal, isobaric, isochoric, adiabatic or follow a more general path.
6. **Forgetting the sign convention for work.** In these notes, expansion means work done by the gas is positive.

16 Summary of key formulas

Concept	Formula
Ideal gas law	$pV = nRT = Nk_B T$
Amount of substance	$n = N/N_A$
Average translational kinetic energy	$\langle K \rangle = \frac{3}{2}k_B T$
Kinetic theory pressure relation	$pV = \frac{1}{3}Nm\langle v^2 \rangle$
RMS speed	$v_{\text{rms}} = \sqrt{3k_B T/m} = \sqrt{3RT/M}$
Monatomic internal energy	$U = \frac{3}{2}nRT$
General ideal gas internal energy	$U = \frac{f}{2}nRT$
First law	$\Delta U = Q - W$
Work in quasistatic process	$W = \int p dV$
Isothermal work	$W = nRT \ln(V_f/V_i)$
Adiabatic relation	$pV^\gamma = \text{constant}$
Heat capacity relation	$C_P - C_V = R$
Sound speed	$c = \sqrt{\gamma RT/M}$

17 Worked examples

Example 5: Heating at constant volume

A monatomic ideal gas has $n = 2.0 \text{ mol}$. It is heated from $T_i = 300 \text{ K}$ to $T_f = 450 \text{ K}$ at constant volume. Find Q , W and ΔU .

At constant volume,

$$W = 0. \quad (71)$$

For a monatomic ideal gas,

$$C_V = \frac{3}{2}R. \quad (72)$$

Thus

$$\Delta U = nC_V \Delta T \quad (73)$$

$$= (2.0 \text{ mol}) \left(\frac{3}{2} 8.314 \text{ J/(mol K)} \right) (150 \text{ K}) \quad (74)$$

$$\approx 3740 \text{ J}. \quad (75)$$

Since $W = 0$, the first law gives

$$Q = \Delta U \approx 3740 \text{ J.} \quad (76)$$

Example 6: Isothermal expansion

One mole of ideal gas expands isothermally at $T = 300 \text{ K}$ from $V_i = 5.0 \text{ L}$ to $V_f = 15.0 \text{ L}$. Find the work done by the gas.

$$W = nRT \ln \left(\frac{V_f}{V_i} \right) \quad (77)$$

$$= (1.0 \text{ mol})(8.314 \text{ J/(mol K)})(300 \text{ K}) \ln 3 \quad (78)$$

$$\approx 2740 \text{ J.} \quad (79)$$

Because the process is isothermal for an ideal gas, $\Delta U = 0$, so $Q = W \approx 2740 \text{ J}$.

Example 7: Adiabatic compression

A diatomic ideal gas with $\gamma = 1.40$ is compressed reversibly from $V_i = 4.0 \text{ L}$ to $V_f = 1.0 \text{ L}$. Its initial temperature is $T_i = 300 \text{ K}$. Find the final temperature.

Using

$$TV^{\gamma-1} = \text{constant}, \quad (80)$$

we get

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}. \quad (81)$$

Therefore

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} \quad (82)$$

$$= 300 \text{ K} \left(\frac{4.0}{1.0} \right)^{0.40} \quad (83)$$

$$\approx 522 \text{ K.} \quad (84)$$

18 Exercises

- Exercise 1.** A gas has volume 3.0 L at pressure 120 kPa . It is compressed isothermally to 1.5 L . Find the final pressure.
- Exercise 2.** A vessel of volume 0.020 m^3 contains an ideal gas at $p = 1.5 \times 10^5 \text{ Pa}$ and $T = 300 \text{ K}$. Find the amount of substance n .
- Exercise 3.** Convert 27°C to kelvin and use it in the ideal gas equation for $n = 0.50 \text{ mol}$, $V = 0.010 \text{ m}^3$. Find the pressure.
- Exercise 4.** For oxygen with molar mass $M = 0.032 \text{ kg/mol}$, find v_{rms} at $T = 300 \text{ K}$.
- Exercise 5.** A monatomic ideal gas with $n = 1.0 \text{ mol}$ is heated from 300 K to 500 K . Find the change in internal energy.
- Exercise 6.** An ideal gas expands at constant pressure $p = 2.0 \times 10^5 \text{ Pa}$ from $V_i = 0.010 \text{ m}^3$ to $V_f = 0.030 \text{ m}^3$. Find the work done by the gas.

- Exercise 7.** One mole of an ideal gas expands isothermally at $T = 400\text{ K}$ from 2.0 L to 8.0 L . Find the work done by the gas.
- Exercise 8.** A gas has $\gamma = 1.40$. During reversible adiabatic expansion its volume doubles. If the initial temperature is 600 K , find the final temperature.
- Exercise 9.** Use $c = \sqrt{\gamma RT/M}$ to estimate the speed of sound in a gas with $\gamma = 1.67$, $M = 0.004\text{ kg/mol}$, and $T = 300\text{ K}$.
- Exercise 10.** Explain in one or two sentences why an ideal gas cools down during reversible adiabatic expansion.

19 Answers to exercises

Answer 1. $p_2 = p_1 V_1/V_2 = (120 \text{ kPa})(3.0/1.5) = 240 \text{ kPa}$.

Answer 2. $n = pV/(RT) = (1.5 \times 10^5 \text{ Pa})(0.020 \text{ m}^3)/[(8.314 \text{ J/(mol K)})(300 \text{ K})] \approx 1.20 \text{ mol}$.

Answer 3. $T = 300.15 \text{ K} \approx 300 \text{ K}$.

$$p = \frac{nRT}{V} = \frac{(0.50)(8.314)(300.15)}{0.010} \approx 1.25 \times 10^5 \text{ Pa}.$$

Answer 4.

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314)(300)}{0.032}} \approx 484 \text{ m/s}.$$

Answer 5.

$$\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}(1.0)(8.314)(200) \approx 2490 \text{ J}.$$

Answer 6.

$$W = p\Delta V = (2.0 \times 10^5 \text{ Pa})(0.020 \text{ m}^3) = 4000 \text{ J}.$$

Answer 7.

$$W = nRT \ln(V_f/V_i) = (1)(8.314)(400) \ln 4 \approx 4610 \text{ J}.$$

Answer 8.

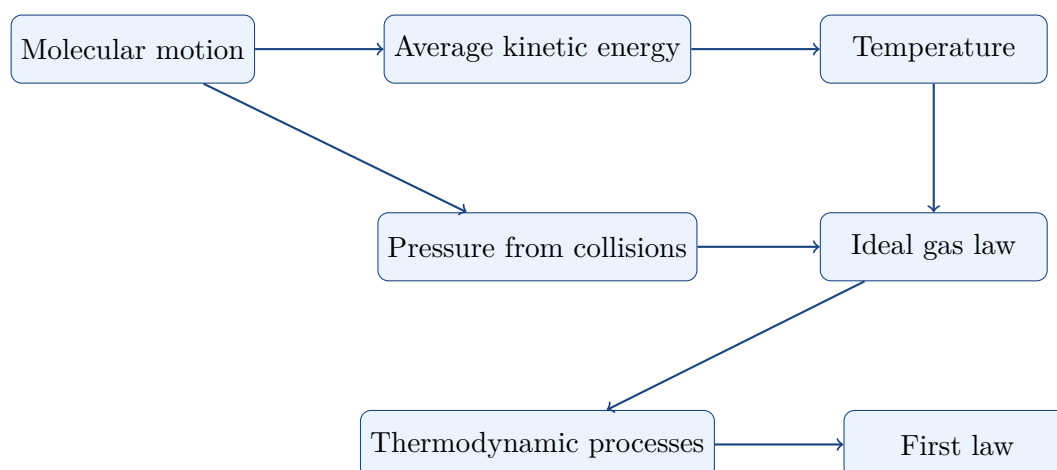
$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} = 600 \left(\frac{1}{2} \right)^{0.40} \approx 455 \text{ K}.$$

Answer 9.

$$c = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{(1.67)(8.314)(300)}{0.004}} \approx 1020 \text{ m/s}.$$

Answer 10. During adiabatic expansion, no heat enters the gas. The gas does positive work on the surroundings, so its internal energy decreases; for an ideal gas this means the temperature decreases.

Concept map



Final remarks

Ideal gas theory is one of the most important bridges between mechanics, thermodynamics and statistical physics. The formula $pV = nRT$ is only the surface of the theory. Behind it lies a powerful idea: macroscopic properties of matter can be explained by the collective motion of microscopic particles.